Co-operative Control through the Lens of Negative Imaginary Systems

Presented by Soorya Krishna
24530011

Systems and Control
Department of Electrical Engineering
Indian Institute of Technology, Roorkee

20th May 2025

Contents

The World of Co-operative Control

In many real-world applications, multiple dynamic systems (agents) coordinate with each other to achieve a collective objective.

- Robotic arms working in sync on a car manufacturing line
- A flock of drones maintaining orientation and formation
- A fleet of autonomous vehicles moving as a platoon
- Smart grids coordinating distributed energy sources

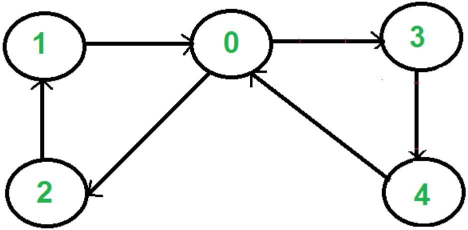
The Consensus Problem

In a network of dynamic systems (agents), **consensus** refers to the process by which all agents agree on a common value, typically a function of their individual states.

A **consensus algorithm** defines how each agent updates its state based on information from neighboring agents.

The communication structure is modeled as a **directed graph**:

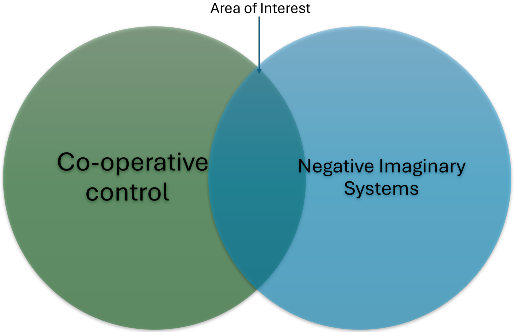
- **Nodes** represent agents (systems)
- **Edges** represent directed information flow or interactions



Pivot Towards Negative Imaginary Systems

Within the broad domain of co-operative control, many real-world systems can be viewed as a special subclass: **Negative Imaginary (NI) systems**.
By leveraging NI system theory, we can design controllers that are both:

- **Robust**, even in the presence of modeling uncertainties
- **Computationally efficient**, ideal for distributed implementations



Introduction to Negative Imaginary Systems

Definition 1: Negative Imaginary (NI) System
Negative Imaginary (NI) systems can be defined as stable systems with a negative imaginary frequency response.
They are mathematically characterized as follows:

$$\mathcal{C} := \left\{ R(s) \in \mathbb{R}\mathcal{H}_{\infty}^{n\times n} : j[R(j\omega) - R(j\omega)^*] \geq 0 \quad \forall \omega \in (0, \infty) \right\}$$

Definition 2: Strictly Negative Imaginary (NI) System
The subclass of **Strictly Negative Imaginary (SNI)** systems is defined as:

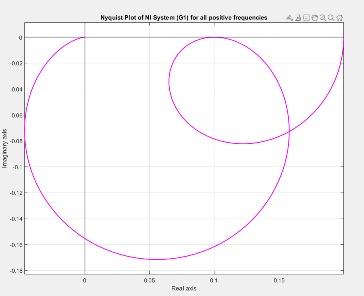
$$\mathcal{C}_s := \left\{ R(s) \in \mathbb{R}\mathcal{H}_{\infty}^{n\times n} : j[R(j\omega) - R(j\omega)^*] > 0 \quad \forall \omega \in (0, \infty) \right\}$$

Negative Imaginary System Examples

Example 1:
Consider the system:

$$M(s) = \frac{(2s^2 + s + 1)}{(s^2 + 2s + 5)(s + 1)(2s + 1)}$$

This is a **Negative Imaginary (NI)** system. Its Nyquist plot *touches the real axis* at $\omega = 0.98$, where the imaginary part vanishes.



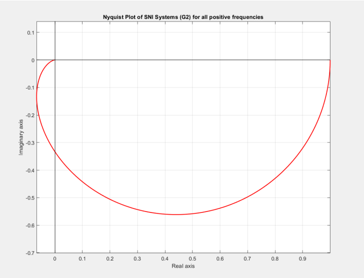
Nyquist Plot of NI System (M(s))

Negative Imaginary System Examples

Example 2:
Consider the system:

$$N(s) = \frac{1}{s^2 + 3s + 1}$$

This is a **Strictly Negative Imaginary (SNI)** system. Its Nyquist plot lies entirely *below the real axis* for positive frequencies.



Nyquist Plot of SNI System (N(s))

Negative Imaginary System Theorem

Background:
Alexander Lanzon and Ian R. Petersen, in their seminal paper **"Stability Robustness of a Feedback Interconnection of Systems With Negative Imaginary Frequency Response"** (IEEE TAC, 2008), developed the following key result:

Theorem 1:
Let $M(s) \in \mathcal{C}$ and $N(s) \in \mathcal{C}_s$ such that:

- $M(\infty)N(\infty) = 0$
- $N(\infty) \geq 0$

Then, the positive feedback interconnection $[M(s), N(s)]$ is internally stable if and only if

$$\bar{\lambda}(M(0)N(0)) < 1$$

where $\bar{\lambda}(\cdot)$ denotes the largest eigenvalue of a matrix.

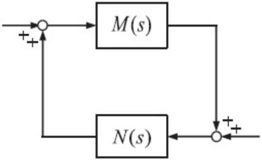


Fig. 1. Positive feedback interconnection.

Importance of the NI Theorem

- Classical tools like the **Nyquist Stability Criterion** provide necessary and sufficient conditions, but require full frequency response data of $M(s)$ and $N(s)$, making them computationally expensive.
- The **Small Gain Theorem (SGT)** requires that the **product of the gains** of $M(s)$ and $N(s)$ remains less than 1 across all frequencies. This condition is often overly conservative for practical systems.
- The **Passivity Theorem** applies only to systems with relative degree ≤ 1, which excludes many real-world co-operative systems involving higher-order dynamics.

Application of NI Theorem

Example 3:

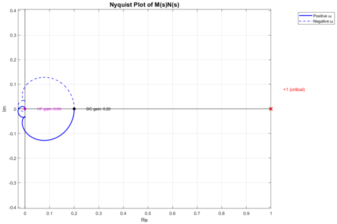
Consider the system pair:

- $M(s) = \dfrac{2s^2 + s + 1}{(s^2 + 2s + 5)(s + 1)(2s + 1)}$ *(NI plant)*
- $N(s) = \dfrac{1}{s^2 + 3s + 1}$ *(SNI controller)*

Observations:

- $M(\infty) = 0, \quad N(\infty) = 0$
- $M(0)N(0) = 0.2 < 1$

Conclusion: The positive feedback interconnection is **internally stable**, which is also confirmed by the Nyquist plot.



Modification of NI Theorem

In 2017, through their paper **"Feedback Stability of Negative Imaginary Systems"**,
A. Lanzon and Hsueh-Ju Chen proposed key modifications to the original NI theorem to address:

- Inclusion of systems with poles at the origin
- Generalization of stability conditions to accommodate a broader class of NI systems

Theorem 2:
Let $M(s)$ be an NI system without poles at the origin and $N(s)$ be an SNI system. Then, the positive feedback interconnection $[M(s), N(s)]$ is internally stable if and only if:

1. $I - M(\infty)N(\infty)$ is nonsingular
2. $\lambda_{\max}\left([I - M(\infty)N(\infty)]^{-1}(M(\infty)N(0) - I)\right) < 0$
3. $\lambda_{\max}\left([I - N(0)M(\infty)]^{-1}(N(0)M(0) - I)\right) < 0$

General NI–SNI Stability Condition

Theorem 3:
Let $M(s)$ be a Negative Imaginary (NI) system and $N(s)$ be a Strictly Negative Imaginary (SNI) system. Let $\Psi < 0$ be a negative definite matrix such that:

$$\bar{\lambda}(M(\infty)\Psi) < 1$$

Then, the positive feedback interconnection $[M(s), N(s)]$ is internally stable if and only if:

1. $I - M(\infty)N(\infty)$ is nonsingular
2. $\bar{\lambda}\left([I - M(\infty)N(\infty)]^{-1}(M(\infty)N(0) - I)\right) < 0$
3. $$\bar{\lambda}\left(\lim_{s\to 0}\left[[I - \Psi M(\infty)][I - N(s)M(\infty)]^{-1}(N(s)M(s) - I)[I - \Psi M(s)]^{-1}\right]\right) < 0$$

SISO Specialisation of Generalised NI Stability Theorem

Corollary 1:

Let $M(s)$ be a scalar Negative Imaginary (NI) system and $N(s)$ a scalar Strictly Negative Imaginary (SNI) system. Suppose $s = 0$ is a (single or double) pole of $M(s)$.

Then the positive feedback interconnection $[M(s), N(s)]$ is **internally stable** if and only if one of the following conditions holds:

1. $M(\infty)N(\infty) < 1, \quad M(\infty)N(0) < 1, \quad \text{and} \quad N(0) < 0$
2. $M(\infty)N(\infty) > 1, \quad M(\infty)N(0) > 1, \quad \text{and} \quad N(0) > 0$

Application of the Generalised NI Stability Theorem

Example 4:
Consider the systems:

- $M(s) = \dfrac{1}{s^2+1} + 2$ (NI system)
- $N(s) = \dfrac{1}{s+1} + 2$ (SNI system)

Standard NI theorem conditions:

- $M(\infty) = 2,\ N(\infty) = 2 \Rightarrow M(\infty)N(\infty) = 4 > 0$
- $M(0) = 3,\ N(0) = 3 \Rightarrow M(0)N(0) = 9 > 1$

The classical NI condition $M(\infty)N(\infty) < 1$ does not hold.
Using the generalised NI theorem:

- $M(\infty)N(\infty) = 4 > 1$
- $M(\infty)N(0) = 6 > 1$
- $N(0) = 3 > 0$

Conclusion: The feedback interconnection is **internally stable** according to the generalised NI result.

Extending Negative Imaginary Systems Theory to Nonlinear Systems

In their 2019 paper titled **"Extending Negative Imaginary Systems Theory to Nonlinear Systems"**,
Ian R. Petersen, Ahmed G. Ghallab, and Mohammed A. Mabrok introduced key definitions that extend NI theory into the nonlinear domain.

Consider the nonlinear system:

$$\dot{x}(t) = f(x(t), u(t)), \quad y(t) = h(x(t))$$

where:

- f is a locally Lipschitz vector field
- h is a continuously differentiable (class $\mathcal{C}^1$) output function

Definition 3: Nonlinear Negative Imaginary (NI) System
The system is said to be *nonlinear NI* if there exists a **positive definite storage function** $V : \mathbb{R}^n \to \mathbb{R}$ such that:

$$\dot{V}(x(t)) \leq u(t)^\top \dot{y}(t)$$

ISNI and OSNI

Definition 4: Input Strictly Negative Imaginary (ISNI) System
The system is said to be *strictly input nonlinear NI* if there exists a **positive definite storage function** $V : \mathbb{R}^n \to \mathbb{R}$ and a scalar $\varepsilon > 0$ such that:

$$\dot{V}(x(t)) \leq u(t)^\top \dot{y}(t) + \varepsilon \|u(t)\|^2$$

Definition 5: Output Strictly Negative Imaginary (OSNI) System
The system is said to be *output strictly NI* if there exists a **positive definite storage function** $V : \mathbb{R}^n \to \mathbb{R}$ and a scalar $\delta > 0$ such that:

$$\dot{V}(x(t)) \leq u(t)^\top \dot{y}(t) - \delta \|\dot{y}(t)\|^2$$

Exploration of Co-operative Control through NI Systems

In their influential paper titled **"Output Feedback Consensus for Networked Heterogeneous Nonlinear Negative-Imaginary Systems with Free Body Motion"**,
Ian R. Petersen, Kanghong Shi, and Igor G. Vladimirov propose a control protocol for achieving output feedback consensus among nonlinear NI systems.

- NI properties naturally emerge in systems such as autonomous vehicles and robotics.
- The agents (nonlinear NI plants) and controllers (nonlinear OSNI systems) are interconnected via a weakly connected directed graph.
- This framework enables robust consensus using only relative output measurements between agents.

Nonlinear NI Feedback Stability Theorem

Theorem 4:
Let:

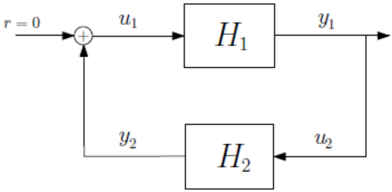
- H_1: a nonlinear negative imaginary (NI) system with state-space model:

$$\dot{x}_1(t) = f_1(x_1(t), u_1(t)), \quad y_1(t) = h_1(x_1(t))$$

- H_2: a nonlinear output strictly negative imaginary (OSNI) system with state-space model:

$$\dot{x}_2(t) = f_2(x_2(t), u_2(t)), \quad y_2(t) = h_2(x_2(t)) + D_2 u_2(t)$$

Then, the positive feedback interconnection of H_1 and H_2 is **asymptotically stable**.

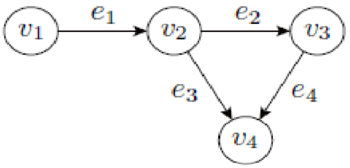


Controller Synthesis Algorithm for Output Feedback Consensus of Nonlinear NI Systems

Consider a multi-agent control system consisting of:

- N negative imaginary (NI) plants represented by the nodes of a graph
- N output strictly negative imaginary (OSNI) controllers represented by the edges

The communication structure among agents is encoded by the **incidence matrix** Q of the underlying directed graph.



Communication Graph

$$Q = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 1 & -1 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Incidence Matrix

Controller Synthesis Algorithm for Output Feedback Consensus of Nonlinear NI Systems

NI Composite Plant H_p:

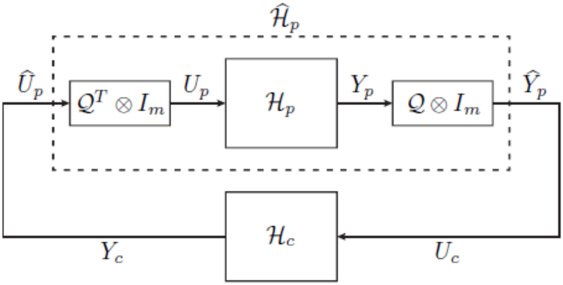
$$\dot{X}_p = \begin{bmatrix} f_{p1}(x_{p1}, u_{p1}) \\ \vdots \\ f_{pN}(x_{pN}, u_{pN}) \end{bmatrix}, \quad Y_p = \begin{bmatrix} h_{p1}(x_{p1}) \\ \vdots \\ h_{pN}(x_{pN}) \end{bmatrix}$$

OSNI Composite Controller H_c:

$$\dot{X}_c = \begin{bmatrix} f_{c1}(x_{c1}, u_{c1}) \\ \vdots \\ f_{cN}(x_{cN}, u_{cN}) \end{bmatrix}, \quad Y_c = \begin{bmatrix} h_{c1}(x_{c1}) \\ \vdots \\ h_{cN}(x_{cN}) \end{bmatrix} + \begin{bmatrix} D_{c1} u_{c1} \\ \vdots \\ D_{cN} u_{cN} \end{bmatrix}$$

Controller Synthesis Algorithm for Output Feedback Consensus of Nonlinear NI Systems

- Each controller processes the difference between outputs of the two plants it connects, capturing disagreement among neighboring agents.
- Each plant receives the sum of outputs from its connected controllers, reflecting the influence of neighboring agents in the network.



Illustrative Example – Output Feedback Consensus in a 4-Agent Network

Example 3:
Consider a network of 4 nonlinear NI agents interconnected via a weakly connected directed graph.

Incidence Matrix Q:

$$Q = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 1 & -1 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Agent (NI plant) dynamics:

$$\dot{x}_{pi}(t) = \mu_i u_{pi}^3, \quad y_{pi}(t) = x_{pi}(t)$$

where μ_i are heterogeneous positive constants.

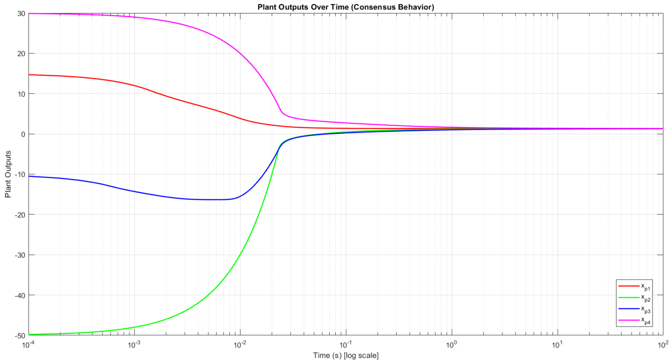
Controller (OSNI) dynamics for each edge:

$$\dot{x}_{ck}(t) = -\alpha_k x_{ck} - \beta_k x_{ck}^3 + u_{ck}(t), \quad y_{ck}(t) = x_{ck}(t) - u_{ck}(t)$$

The goal is to achieve output consensus using only relative output feedback through the directed communication graph.

Simulation Results – State Trajectories

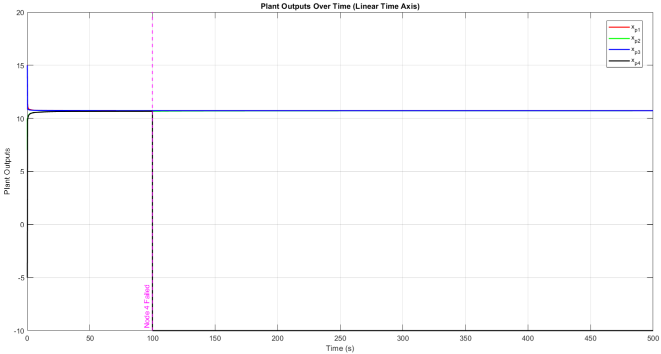
The following plot shows the state trajectories of the 4 nonlinear NI agents over time. Each agent's state evolves under the influence of the OSNI controllers using only relative output feedback. Note - Time is in logscale.



All agents converge to a common trajectory, demonstrating output consensus.

Simulation Results – Consensus under failure of a node

After achieving consensus, node 4 undergoes a failure and drops out.



Despite the failure, the systems maintain consensus and remains unaffected.

Conclusion

- Reviewed the concept of Negative Imaginary (NI) systems and presented key theorems ensuring internal stability for positive feedback interconnections of NI and SNI systems.
- Explored the extension of NI theory to the nonlinear domain, enabling its application to systems with nonlinear dynamics and free body motion.
- Highlighted that many real-world multi-agent systems—such as robotic , UAV , and vehicular platoons—naturally exhibit nonlinear NI properties.
- Showcased how controller synthesis based on NI theory enables robust, decentralized, and computationally efficient output feedback control for cooperative systems.

References

1. A. Lanzon and I. R. Petersen,
"Stability Robustness of a Feedback Interconnection of Systems With Negative Imaginary Frequency Response,"
IEEE Transactions on Automatic Control, vol. 53, no. 4, 2008.
2. A. Lanzon and H.-J. Chen,
"Feedback Stability of Negative Imaginary Systems,"
IEEE Transactions on Automatic Control, vol. 62, no. 11, 2017.
3. I. R. Petersen, A. G. Ghallab, and M. A. Mabrok,
"Extending Negative Imaginary Systems Theory to Nonlinear Systems,"
Proceedings of the 2018 IEEE Conference on Decision and Control (CDC).
4. I. R. Petersen, K. Shi, and I. G. Vladimirov,
"Output Feedback Consensus for Networked Heterogeneous Nonlinear Negative-Imaginary Systems with Free Body Motion,"
IEEE Transactions on Automatic Control, vol. 68, no. 9, 2023.
5. M. A. Mabrok, A. G. Kallapur, I. R. Petersen, and A. Lanzon,
"Generalizing Negative Imaginary Systems Theory to Include Free Body Dynamics: Control of Highly Resonant Structures with Free Body Motion,"
IEEE Transactions on Automatic Control, vol. 59, no. 10, 2014.